

Clean E13-family implementation and Serv runbook

- **Date:** 10 August 2026
- **Branch:** kit/e13-family-clean
- **Clean base:** 2157eada14824d14019e80f9416e6d736c837306 (code clean-up - restore 1 ref only, 2 June 2026 21:58 BST)
- **Supported recipes:** E13, BC_E13, and CL14
- **State:** implemented, locally audited, committed in the branch history, and deliberately not launched

Outcome

The branch keeps the June single-reference model as its base and adds one small, fail-closed E13-family contract. E13 and BC_E13 share exactly the same model, optimizer, pipeline, schedule, and validation behavior; their only difference is the training dataset. CL14 uses the same contract with two named training-side changes: the inherited CL9 Cosmic dataset policy and a two-latent- cell target-mask feather. CL14 does not change the inference mask or denoising equation.

The implementation intentionally separates four concerns:

```
thin leaf config
  shared hard-v1 E13 model/checkpoint contract
  sealed CL14 inference/validation path
  one dataset policy (Large, BigCelebs, or Cosmic)
  explicit training-efficiency profile
```

Later E14-E24 mechanisms were not copied. In particular, there is no residual, anchored, or query-adaptive BA; no identity cross-attention; no ArcFace auxiliary; no reference dropout; no ROI warp; and no branch-output adapter.

Provenance

The implementation was reconstructed path-by-path rather than by merging `test`. The source hierarchy was:

Purpose	Immutable source
June clean base	2157eada14824d14019e80f9416e6d736c837306
strict E13 substrate	e860f9ed4d021226575845ae24a9fda1e5a3fa58
effective adapter scopes/rank	dd65dec312271610e46dd507ee010a5927b8cbda
joint groups/validation shadow	8b8b9abd726df111ce725b6c283869c3dd19e6a0
successful E13 runtime	ebf1ac8295f363adb0055cd74db1a96c2ff03a35
BC_E13 runtime	ad194a026ab701dd979712d415c487dd536a4645
exact CL14 overlay	c04970f342a186d1092f07f9a08d7d8a797383e8+c112-c114-snapshot-v1-20260809

Sealed CL14 evidence:

- source-manifest SHA-256: d43fa65815aa4fc4c106f6ed3e939b5dee690f2a2927b43a022a8e6025ccc294;
- resolved saved-config SHA-256: 642cdcb4acd2b4fcf0ef9fd5fadaa5fb5a117c092b1e07394b8e9c2fd406b2c4;
- 24k weights SHA-256: 0de10ec611c8a5e55e0b362ea90fa348fe686ca5d949f762752bc1add7992ed9;
- immutable Comet experiment key: 6fe0028be92242c38056b3d36665fdd6.

Other historical records:

Run	Comet key	Known 24k checkpoint SHA-256
E13 r4	1cc0a02371094b24a6a02a4cc649f10c	4a9d95a3f957609fcf4eb77771f263dec8e71189dc72aae347233091de4249ab

Run	Comet key	Known 24k checkpoint SHA-256
BC_E13 r1	c138db7c41ae435c8a7560f40cf5f58d	99b305bad425dd07073a4a54e0a978dea0d4a02456c8129eb1b12afb5a459e
CL14 r1	6fe0028be92242c38056b3d36665fdd6	0de10ec611c8a5e55e0b362ea90fa348fe686ca5d949f762752bc1add7992ed9

BigCelebs sealed-v2 manifest SHA-256 is f846b8cc8a4ce087c78130beee48a65f1b13560b63e42a9715cb5686526e5efa. The historical E13 Large Dataset manifest SHA-256 is 0056f9647c6ca69079c3b7ae479ea5cdf9e642f076460249b160000eecb3ee50.

Supported recipe matrix

Field	E13	BC_E13	CL14
leaf config	E13_large_ds_joint_shadow_sa128_24k	BC_E13_big_celebs_joint_shadow_sa128_24k	CL14_cosmic_joint_shadow_sa128_softmask_24k
dataset	LargeDatasetTrain	BigCelebsTrain	CosmicLargeAdaptedTrain
model architecture	hard-replace-v1	identical	identical
target training-mask	0	0	2
feather			
inference mask	hard historical mask	identical	identical
fixed-96 active bbox	E13/BC 4db6344d...	identical	CL10-CL14 b33cf026...
cache			.
optimizer ownership	840/700/700	identical	identical
schedule/validation/runtimeshared		shared	shared

All leaves resolve to 24,000 optimizer steps: batch size 2, 2,000-step epochs, 12 epochs, checkpoint and fixed-96 validation at step 0/every 2,000 steps, and WarmupHoldCosineLR(20, 14000, 24000, 0.1). Validation uses DDIM 50, CFG 5, PhotoMaker from step 10, and BA from step 15. Eligible BA invariants remain `pose_adapt_ratio=0` and `ca_mixing_for_face=false`.

Change ledger

Every new logical code block is adjacent to a dated **E13C-*** marker. Existing AICODE-NOTE: anchors were retained where they document critical cache, routing, dataset, or deferred-scoring invariants.

Core model and checkpoint

- E13C-CORE-01 (`lora2.py`, `lora2_helpers.py`, `branched_runtime.py`): strict hard-v1 installation and route validation prevents silent fallback to a plain processor or wrong architecture.
- E13C-CORE-02 (`e13_contract.py`, `branched_runtime.py`): the rank-128 hard branch preserves generic rank 32 in the historical manifest, reproducing E13 capacity and the old checkpoint schema.
- E13C-CORE-03 (`e13_contract.py`, `lora2.py`, `train.py`): the exact 840/700/700 allowlist and optimizer grouping prove every trained tensor has one owner and no base U-Net tensor trains.
- E13C-CORE-04 (`e13_contract.py`, `lora2.py`): complete schema-v2 state, shapes/dtypes, semantic processor hash, and strict loading save all three trained paths and reject incompatible checkpoints.
- E13C-CORE-05 (`base_trainer.py`): restoring 700 pretrained PhotoMaker-default tensors and full-copying 70 validation processor states reproduces E13 shadow-coadapter validation without changing saved weights.
- E13C-CORE-06 (`lora2.py`): CL14's 1/3, 2/3, 1 inward target-mask ramp teaches a gradual training handover while leaving reference and inference masks unchanged.

Pipeline and validation

- E13C-PIPE-01 (`photomaker_branched_clean.py`, `br_pipeline_helpers.py`): one spatial reference per 12-prompt identity batch prevents accidental repeated or multiple spatial lanes.
- E13C-PIPE-02 (pipeline files, `branched_runtime.py`): sealed reference latent/mask/noise preparation and denoising preserve CL14 RNG consumption and generation equations.
- E13C-PIPE-03 (leaf configs, fixed-96 protocol files, validators): pinned manual and run-family-specific automatic caches prevent face-detector or bbox-map drift from changing validation generations.

Datasets

- E13C-DATA-01 (`large_dataset.py`, registry): distinct same-ID target/reference loading preserves the exact E13 data relationship.
- E13C-DATA-02 (`big_celebs.py`, registry): sealed field, bbox, trigger, and minimum-face validation makes BC_E13 a dataset-only transfer and fails early on a bad release.
- E13C-DATA-03 (`cosmic_large_adapted.py`, `reference_policy.py`): disabling reference flips and using pose-first, one-trigger, 50-word prompts avoids independent identity mirroring and loss of pose/background tokens.
- E13C-DATA-04 (`reference_frame.py`, Cosmic loader): a 1024 target-frame reference with 6%-30% face area and 0.15 position jitter fixes the 256-pixel tight-crop scale mismatch without a fixed-position shortcut.

Training efficiency and operations

- E13C-PERF-01 (`lora2.py`, `lora2_helpers.py`): batched frozen text/PhotoMaker/VAE conditioning and the cache-off diverse-pair policy remove repeated per-sample frozen encoder work.
- E13C-PERF-02 (model/runtime/trainer): skipping unreachable text-only work and host sync, caching masks, disabling unused debug output/zero-touch, and sparse grad telemetry remove output-irrelevant work while preserving June defaults outside the family.
- E13C-PERF-03 (active/Serv launchers, model preflight): asynchronous CUDA and exact ORT-GPU 1.20.1 fail-closed checks prevent the prior silent 5-7 seconds/step CPU or blocking path.
- E13C-PERF-04 (trainer, face-quality module, Comet tools): staging images and scoring after successful training prevents PyIQA from delaying, perturbing, or invalidating optimizer/checkpoint work.
- E13C-CFG-01 (shared config, scheduler): one architecture, schedule, and efficiency contract makes cross-dataset equivalence reviewable.
- E13C-CFG-02 (three leaves, launcher, validators): only dataset and CL14 mask deltas live in leaf configs; required hashes reject ad-hoc drift before GPU startup.
- E13C-DOC-01 (this file, handoff, tools index): complete provenance, evidence, limitations, and runbook let another agent audit and run the branch without reconstructing its history.

1. Core model and checkpoint implementation

`e13_contract.py` is the only E13-specific ownership switchboard. It freezes everything, then enables exactly:

Optimizer group	Tensors	Parameters	LR
rank-128 hard spatial BA	840	127,795,200	1e-4
rank-32 generic effective adapter	700	30,474,240	1e-4
rank-64 PhotoMaker-default effective adapter	700	60,948,480	1e-4
total	2,240	219,217,920	—

The branch rank is stored as `hard_v1_extensions.lora_rank=128`; the legacy top-level `branched_attn_lora_rank=32` remains the generic rank exactly as in the sealed schema. The loader compares the output-affecting E13 projection, semantic processor-name hash, exact trainable names, shapes, and dtypes. It allows unrelated later manifest fields so genuine E13/BC_E13/CL14 schema-v2 checkpoints remain loadable, but rejects route, scope, timing, rank, processor, name, or shape drift.

The current full construction produced semantic processor hash `1c34710dd22f0108d015d1ac0d62c1d4a23ae271853fdfdb26c58d05ac717e3d`.

2. Pipeline and CL14 generation parity

The two pipeline files are the sealed CL14 files plus only a leading dated audit marker. After removing that marker and trailing blank lines, SHA-256 is:

File	Sealed canonical SHA-256
src/pipelines/br_pipeline_helpers.py	4c1516d3536a85c028580f601b61773df55c49d6b16d fac9d93c997102be5c95
src/pipelines/photomaker_branched_clean.py	85e1b3a2da90ba4a007f8bda895c722c7a21c8e5a519 b86881626ed665e9071c

The complete AST source for `two_branch_predict`, canonicalized only for historical trailing spaces, hashes to 9145856534abe92a6f48e9328dad5a1692ff65f27a51ac3554f9b4db82ad3689. `tools/verify_cl14_generation_parity.py` checks all three values before every server launch.

Generation-box routing is also pinned because it is an input to the denoising mask. Both families use the 96-entry canonical manual protocol `a39645e22b68027175946a028e185b7c5393a7514f5d68c94cd74e7cc9f5e614`, whose `Reading pa_jensen.png` entry is intentionally `force_manual`. E13 and BC_E13 load the historical 96-entry automatic cache `4db6344d0deb0af0ee7a25d839b774c9a4a0c5b8f6ff4cc00aaa9c0d6d85c099`; CL14 loads the later CL10-CL14 cache `b33cf02665cd875a738fba8f20c2ea95fcb0585358436e32691af0013a2f1c7d`. Thus Jensen uses its manual record and the other 95 routes use the selected cache. The files live under `dataset_full/val_dataset/protocols/`, and both validators reject a missing, renamed, reserialized, or regenerated map before model loading. The parity gate also seals the prompt file, class map, reference boxes, legacy identity embeddings, and all 12 reference images byte-for-byte.

The attention processor is intentionally smaller than the sealed post-CL14 switchboard. Under CL14’s selected flags, a deterministic fixture with nonzero branch-LoRA weights produced `torch.equal=True` against the sealed processor source. The only active addition is per-forward mask memoization; the cached tensor is the same resized mask object and cannot cross samples or timesteps.

These facts establish source-level pipeline parity and bit-exact selected-processor parity. A fresh historical-checkpoint/full-96 RGB replay was **not** run locally: the available machine is not the pinned A100/ORT-GPU runtime, the historical weight artifact is not present locally, and the user explicitly asked not to launch the Serv job yet. Consequently this document does not claim a newly measured pixel replay. The first Serv run must retain the generated step-0 panel and compare it with the immutable CL14 export before promotion.

CL14’s mask feather cannot affect inference: it is read only by training `_bbox_to_mask`; the reference mask and the pipeline’s inference mask are unchanged.

3. Dataset improvements

Dataset changes finish before model input and do not appear in model or pipeline conditionals.

E13 Large Dataset

The loader samples a distinct image from the same identity, propagates target horizontal flips to its bbox, never independently flips the reference, and emits the final reference image, bbox, identity, path, and cache key.

BC_E13 BigCelebs

BC_E13 inherits E13 and changes only `train_dataset_name`. The loader accepts the sealed manifest schema (`new_face_crop`, `text`), requires a minimum 192-pixel face side, exactly one `img` trigger, and at least two images per identity. The launcher hard-pins the sealed-v2 manifest hash.

CL14 Cosmic

The fixes are separate and explicit:

1. Reference geometry: the 256-pixel crop is composed into a 1024 target frame before the VAE; face area is sampled from 6%-30% and position jitter is 0.15.
2. Reference hygiene: independent reference mirroring is disabled.
3. Prompt hygiene: pose-first captions preserve one lowercase `img` trigger and cap at 50 words so pose/background content normally survives tokenization.
4. Cache correctness: target-dependent scale/position framing is encoded in the reference cache key.

The launcher requires the exact Cosmic manifest hash to be recorded in `.env`. There is no known hash hard-coded because this repository snapshot did not contain the data artifact; the operator must take it from the sealed CL14 data package rather than substituting a similarly named manifest.

4. Training-efficiency improvements

Efficiency is an explicit profile, not an architecture or dataset claim:

Switch	Selected value	Work avoided	Numerical status
<code>conditioning_cache_enabled</code>	false	ineffective diverse-pair cache bookkeeping	output-neutral
<code>batched_conditioning_preparation</code>	true	per-sample frozen text/ID/VAE calls	historical E13-family path; semantically equivalent, not bit-identical to unbatched bf16 GEMMs
<code>skip_unused_text_conditioning</code>	true	unreachable text-only encode and <code>timestep.item()</code> sync	output-neutral because BA trains at every timestep
<code>cache_prepared_masks</code>	true	repeated mask resize in one doubled forward	exact tensor reuse
<code>compute_branch_debug_outputs</code>	false	unused post-merge diagnostic tensors	output-neutral
<code>post_backward_parameter_touch</code>	false	full zero-valued trainable scan	cannot affect optimizer values
<code>grad_norm_log_only</code>	true	norm reductions outside logging steps	telemetry-only
<code>train workers</code>	2	<code> DataLoader stalls</code>	same historical setting; persistent workers remain off

Historical evidence measured roughly 5 s/step falling to roughly 0.9 s/step after batched conditioning. One bf16 step differed by about 0.074% in loss from the scalar implementation, so it is deliberately described as the exact run setting—not as bit-identical to the slower path.

Production also requires unset/zero `CUDA_LAUNCH_BLOCKING`, ONNX Runtime GPU 1.20.1, an available and actually loaded CUDA provider in every InsightFace session, and single-process Accelerate. Failure is fatal before training.

Face-quality generation remains step 0/every 2,000; only PyIQA scoring moves after successful training. The finalizer checks the immutable Comet key, all 13 expected steps, 96 images per step, and per-image assets.

5. Concision review

The model was not copied wholesale from the 2,335-line sealed experimental switchboard:

File	Clean branch lines	Sealed CL14 lines
<code>lora2.py</code>	953	2,335
<code>lora2_helpers.py</code>	505	949

File	Clean branch lines	Sealed CL14 lines
<code>branched_runtime.py</code>	680	1,214
<code>attn_processor_cleanest.py</code>	779	931
<code>new focused e13_contract.py</code>	311	not separate

The pipeline is deliberately not shortened because exact CL14 source parity is more important than cosmetic cleanup: its two files are sealed content plus one audit comment. Rich dataset and Comet/face-quality tools remain outside the model and do not enlarge its architecture switchboard.

Retained standalone tooling includes immutable-key Comet retrieval/export, deferred and historical face-quality scoring, the three dataset preflights, face/body alignment measurement, the research-report skill and renderer, and the hash-verifying Dropbox uploader. `TOOLS.md` lists only utilities actually present on this branch; later architecture evaluators and Serv runtime debris were intentionally not copied.

6. Verification performed

Gate	Result
branch/base	clean worktree created directly from exact 2 June SHA; no merge from <code>test</code>
changed Python compilation/import	passed
launcher shell syntax	passed
Hydra composition	all three recipes passed <code>validate_e13_family_config.py</code>
ownership	full SDXL CPU construction passed exact 840/700/700 and 2,240/219,217,920 counts
manifest	generic rank 32, hard rank 128, 2,240 names, semantic hash present
schema-v2	save/load round trip passed; wrong semantic processor hash rejected
pipeline	both canonical sealed file hashes passed
denoising	sealed <code>two_branch_predict</code> AST hash passed
fixed-96 inputs	prompt/classes/reference-box/embedding/12-image hashes passed; canonical manual and separate active-cache hashes passed; 96 records and Jensen override verified
processor	deterministic nonzero-LoRA hard-v1 fixture was bit-exact to sealed CL14 source
mask	E13/BC binary; CL14 rings exactly 1/3, 2/3, 1
datasets	distinct Large/BigCelebs refs, strict BigCelebs, Cosmic prompt and target-frame scale ratio 1.0 passed
deferred scoring	staged two images and immutable experiment key without importing/running PyIQA
Comet contract	logger writes <code>saved/<run>/comet_experiment.json</code> ; launcher requires it after training
diff hygiene	focused <code>git diff --check</code> passed before commit

A repository-wide `compileall` is not a valid clean signal on this June base: the pre-existing tracked file `src/model/attn_procs/attn_processor.py` starts with invalid `phaimport torch` in both the base commit and this branch. It was not changed because it is outside this implementation and is not the selected processor. Focused compilation covers every changed Python file.

Not performed yet:

- no A100 benchmark or training job;

- no historical weight download/checkpoint load;
- no fixed-96 RGB replay or metric recomputation;
- no push to a remote branch.

7. Serv preparation and launch instructions

Do not run these commands until the branch is intentionally transferred to the Serv checkout and a job launch is approved. Never use Neb; it is unavailable.

7.1 Pin the source

After this local branch is pushed or otherwise transferred, use a dedicated Serv checkout/worktree and pin the final commit shown by local `git log`:

```
git fetch origin kit/e13-family-clean
git switch kit/e13-family-clean
git pull --ff-only
git rev-parse HEAD
git status --short
```

The status must be empty. Record that commit in the run notes. Do not train from an uncommitted NFS copy and do not add Hydra overrides.

7.2 Configure machine-local inputs

Run from `diffusion_template/` and keep secrets/paths in the ignored `.env`:

```
cp .env.example .env
chmod 600 .env
sha256sum /path/to/large_manifest.json
sha256sum /path/to/cosmic_manifest.json
```

Fill only the recipe-relevant dataset variables plus:

- `COMET_API_KEY`;
- `FACE_QUALITY_SCORER_PYTHON`, pointing to the existing PyIQA 0.1.15 interpreter;
- `PM_PATH` if the pinned PhotoMaker v2 file is not at the cached default;
- the exact dataset manifest path, image/root path, and manifest SHA-256.

For Large Dataset and BigCelebs, do not change the supplied historical/sealed hashes. For Cosmic, copy the hash from the exact historical CL14 data package or immutable run record and record it in the run notes. Do not commit `.env`.

7.3 Run read-only preflights

Use the existing Serv `photomaker_NS` environment:

```
cd /absolute/path/to/diffusion_template
conda activate /absolute/path/to/conda_env/photomaker_NS
python tools/validate_e13_family_config.py
python tools/verify_cl14_generation_parity.py
bash -n launchers/active/run_e13_family_24k_1gpu.sh
bash -n launchers/serv/start_e13_family_1gpu.sh
```

The active launcher repeats the config/parity checks and then requires ORT-GPU 1.20.1/CUDA provider activation before model construction.

7.4 Prepare one MLS request

Copy `serv_run_packages/e13_family_1gpu.yaml.example` to a run-specific path and replace every `REPLACE_*` value:

- project root is the absolute `diffusion_template` checkout;
- Conda path is the existing Serv `photomaker_NS` environment;
- log directory and run name are unique;
- config is exactly one supported leaf name from the recipe matrix.

Before submission, inspect both queues:

```
mls job list --status Running --limit 100 --output json
mls job list --status Pending --limit 100 --output json
```

Count this project's one- and two-GPU Running/Pending requests. The normal ceiling is six requested A100s. Do not submit if the new one-GPU request would exceed it. The current task authorizes preparation only, not submission.

When separately approved, the submission command is:

```
mls job submit --config /absolute/path/to/run_e13_family_1gpu.yaml
```

7.5 Startup and completion checks

Monitor the run-specific stdout/stderr paths from the YAML. Startup is valid only after:

- the manifest SHA check passes;
- the run-specific dataset decode/policy JSON exists under `preflight_records/<RUN_NAME>/`;
- config/parity validators pass;
- the fixed-96 manual and recipe-specific automatic bbox hashes pass, so no detector pass can replace the historical generation-mask inputs;
- ORT reports 1.20.1 with `CUDAExecutionProvider`;
- ownership logs 840/700/700 and the total 2,240/219,217,920;
- `saved/<RUN_NAME>/comet_experiment.json` exists and contains the immutable experiment key.

Expected validation/checkpoint boundaries are 0,2000,...,24000. After successful training, the same launcher invokes the deferred face-quality finalizer. Scoring failure is nonfatal to completed model artifacts but must be repaired before the run is called fully reported.

For CL14, preserve the step-0 full-96 images before continuing promotion work and compare them against the immutable historical export. Any mismatch should be investigated at the first divergent tensor/image; do not replace pixel parity with aggregate metric similarity.

Related documents

- [Implementation plan](#)
- [Current handoff](#)
- [Validation protocol](#)
- [Tool index](#)